UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

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|---|---|
| COURSE | INTRODUCTION TO INTERACTIVE MULTIMEDIA |
| COURSE CODE | CSC413 |
| EXAMINATION | NOVEMBER 2026 |
| TIME | 3 HOURS |

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

*This examination paper consists of 5 printed pages.*

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## PART A (15 MARKS)

*State TRUE (T) or FALSE (F) for the following statements.*

1. A storyboard is a visual plan of a multimedia sequence.
2. Haptic feedback provides tactile sensations.
3. Cel animation is created by drawing each frame by hand.
4. Hypermedia combines multimedia with hyperlinks.
5. Multimedia combines text, graphics, audio, video and animation.
6. Interactivity gives users control over content and flow.
7. 3D animation does not use modelling.
8. A graphics tablet is used to draw directly into a computer.
9. MPEG is a common family of video compression formats.
10. Keyframes define important positions in an animation.
11. A low-fidelity prototype is used to test the basic layout and flow.
12. A prototype is an early model of a product.
13. Tweening creates intermediate frames between keyframes.
14. CMYK is used for printed output.
15. A multimedia project always uses every media element.

## PART B (15 MARKS)

*Select the most appropriate answer from Table A for each of the following questions.*

*Table A. Matching options*

| | | |
|---|---|---|
| A. Keyframe | B. Frame rate | C. HTML |
| D. Hertz | E. Browser | F. RGB |
| G. Sampling | H. Storyboard | I. Vector |
| J. MIDI | K. Scanner | L. Bitmap |
| M. Copyright | N. Hyperlink | O. Amplitude |

1. image made of pixels
2. visual sequence plan
3. converting sound into digital values
4. image made of mathematical objects
5. standard for musical notes and instructions
6. software that displays web pages
7. language for web structure
8. input device that digitises documents
9. frames displayed per second
10. height of a sound wave
11. important frame in an animation
12. additive colour model used by screens
13. clickable link to another resource
14. exclusive rights of a creator
15. unit of frequency

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## PART C (30 MARKS)

*Choose the correct answer for each question.*

1. Plug and play means:
   A. No power needed
   B. Automatic device recognition
   C. Wireless only
   D. Manual driver installation only

2. A higher sampling rate generally:
   A. Improves accuracy
   B. Removes audio
   C. Changes colour
   D. Reduces accuracy

3. Which is an output device?
   A. Scanner
   B. Mouse
   C. Plotter
   D. Keyboard

4. Path animation moves an object:
   A. Without motion
   B. Randomly
   C. Along a predefined route
   D. Only vertically

5. Frame rate is:
   A. Frames per second
   B. Bits per second
   C. Samples per second
   D. Pixels per inch

6. Which image type scales without losing quality?
   A. Vector
   B. Bitmap
   C. Scan
   D. Photograph

7. Which colour model is subtractive?
   A. CMYK
   B. HSL
   C. HSV
   D. RGB

8. Streaming allows media to:
   A. Require full download first
   B. Play while downloading
   C. Only play offline
   D. Remove compression

9. Which is a benefit of multimedia in education?
   A. It combines visuals and sound
   B. It uses only text
   C. It replaces all teachers
   D. It removes all interaction

10. RAM is:
    A. Temporary working memory
    B. An input device
    C. A web browser
    D. Permanent storage

11. An authoring tool is used to:
    A. Only browse the web
    B. Assemble multimedia elements
    C. Only print text
    D. Only record audio

12. Chroma key is used to:
    A. Replace a colour background
    B. Compress audio
    C. Add text
    D. Scan images

13. Which is an example of linear multimedia?
    A. An interactive kiosk
    B. A game with choices
    C. A website with menus
    D. A cinema presentation

14. Kerning adjusts:
A. Line length
B. Space between individual character pairs
C. Font colour
D. Page size

15. A presentation where the user controls the sequence is called:
A. Printed
B. Static
C. Non-linear
D. Linear

16. Anti-aliasing is used to:
A. Smooth jagged edges
B. Add sound
C. Compress video
D. Increase file size only

17. Which is a 3D animation task?
A. Modelling and rendering
B. Bookmarking
C. Kerning
D. Sampling

18. Which is an input device?
A. Printer
B. Scanner
C. Monitor
D. Plotter

19. Card-based authoring tools are suited to:
A. Audio editing only
B. Live video only
C. Messages with a beginning and end
D. 3D modelling only

20. A prototype is:
A. An early model of a product
B. A codec
C. A font
D. A final release

21. Hypertext contains:
A. Links to other content
B. No text
C. Only images
D. Only audio

22. A storyboard is:
A. A video codec
B. A visual sequence plan
C. A sound file
D. A colour model

23. Hypermedia is best described as:
A. A single video file
B. Multimedia with hyperlinks
C. A printed manual
D. Text-only media

24. A bitmap image is made of:
A. Mathematical objects
B. Sound samples
C. Pixels
D. Text characters

25. Which best defines multimedia?
A. A single image
B. A printed book
C. Text only
D. Text, graphics, audio, video and animation

26. MPEG is:
A. A font
B. A video compression family
C. A storage device
D. An image editor

27. A hyperlink links to:
A. Another resource
B. Only images
C. Only audio
D. Only text

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28. A serif font has:

A. Only uppercase letters
B. No spacing
C. Small strokes at character ends
D. No strokes

29. Which is an audio file format?

A. WAV
B. HTML
C. MPEG
D. JPEG

30. Convergence in multimedia refers to:

A. Merging of media and computing
B. Recording audio only
C. Deleting media files
D. Printing images

## PART D (40 MARKS)

### QUESTION 1

A shopping complex requires a kiosk that gives customers location and retailer information.

**a)** State TWO important project team roles for completing the kiosk.

*(2 marks)*

**b)** Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

*(3 marks)*

**c)** Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

*(5 marks)*

### QUESTION 2

A company wants an interactive product demonstration that runs on a kiosk and on the web.

**a)** State TWO features to consider when choosing authoring software.

*(3 marks)*

**b)** Explain the interactivity level suitable for the demonstration.

*(3 marks)*

**c)** Explain TWO authoring tool categories appropriate for this project.

*(4 marks)*

### QUESTION 3

A museum wants a website that presents its collection and lets visitors explore related exhibits.

**a)** State TWO differences between linear and non-linear navigation.

*(3 marks)*

**b)** Describe TWO roles of hyperlinks in the website.

*(3 marks)*

**c)** Explain how streaming and compression affect multimedia delivery on the web.

*(4 marks)*

### QUESTION 4

A product catalogue must show sharp images on screen and in print.

**a)** State TWO differences between bitmap and vector images.

*(2 marks)*

**b)** Describe TWO factors that affect image quality.

*(4 marks)*

**c)** Explain how colour model and resolution affect screen and print output.

*(4 marks)*

*END OF QUESTION PAPER*